

Exiled Regime and Gender Norms Diffusion

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Abstract

This paper examines the horizontal transmission of gender attitudes across ethnic groups in Taiwan. Following the *Kuomintang's* retreat to Taiwan after the Chinese Civil War, the government constructed a series of public settlements (*Juancun*) to accommodate the incoming mainland migrants. Using the precise locations of *Juancun*, I quantify the intensity of local exposure and show that greater exposure to gender-egalitarian migrant communities led to significant shifts in native Taiwanese gender outcomes, including higher female labor force participation, lower fertility and a weakened preference for sons.

Keywords: exiled regime; cultural diffusion; gender norms; female labor force participation; son preference.

JEL: J16; O15; N35.

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1 Introduction

Culture plays a central role in shaping economic development and institutional formation. This paper builds on this insight by examining how the exchange of cultural values between different ethnic groups affects economic outcomes. Identifying the causal effects of cultural transmission, however, is empirically difficult due to endogeneity concerns such as omitted variable bias and measurement error. To overcome these challenges, a growing literature exploits historical episodes of forced migration as exogenous shocks ([Becker et al. \(2020\)](#); [Miho et al. \(2023\)](#)) to study how exposure to different ethnic groups influences local economic and attitudinal outcomes.

This paper exploits a unique historical event—the mass migration following the Chinese Civil War between 1945 and 1950—to examine how the large inflow of Chinese mainland immigrants influenced local Taiwanese economic outcomes, particularly those related to gender. This historical episode offers an ideal setting to study horizontal cultural transmission for two key reasons. First, mainland immigrants were, on average, more gender-egalitarian than local Taiwanese, as evidenced by a smaller gender gap in employment and weaker son preference. Second, local variation in exposure to mainland immigrants was plausibly quasi-exogenous.

After the Chinese Civil War, according to [Yap \(2021\)](#), around a million of Chinese Mainland surged into Taiwan, causing a enormous population shock to the Taiwanese society. In order to accommodate these migrants, the government built a great number of public settlements, also known as *Juancan*. Anecdotal evidence suggests the sites to build Juancan were mainly chosen by administrative consideration instead of gender related reasons. Unlike prisons or other forms of forced housing, Juancan lacked clear physical barriers or boundaries, which facilitated interactions between residents and the local population.

I examine whether gender-related outcomes among local Taiwanese were shifted by the exposure to Juancan. Specifically, I combine historical micro-level census data with detailed geographic information on the distribution of Juancan to construct an index of exposure intensity. Relative to existing studies, the data used in this paper provide substantially more granular residential

information-at the village level-including the exact locations of individual Juancan. This feature allows me to exploit individual-level variation in distance to nearby Juancan, enabling a more precise measure of exposure intensity. This paper also contributes to the literature on cultural transmission and economic development by examining a unique historical migration event in which the identity of immigrants differs remarkably from those studied in the existing forced migration literature. In contrast to prior studies that primarily focus on displaced or marginalized groups-such as refugees or exiled populations-the migrants in this setting were predominantly urban elites, military soldiers, and their families. These groups played a central role in national development, and thus carried distinct social norms and economic attitudes. This feature allows the paper to shed new light on how cultural transmission operates when migrants are not socially or economically disadvantaged, but instead occupy influential positions in the host society.

The findings suggest significant shifts in gender-related outcomes. As exposure increases, local Taiwanese women become more likely to be employed, have lower fertility, and, more importantly, exhibit weaker son preference. In contrast, there is no evidence of positive effects on men’s employment outcomes. Taken together, greater exposure to more gender-egalitarian groups appears to have altered intra-household divisions of labor and the associated fertility decisions among local Taiwanese households. These results are robust across a wide range of specifications and alternative empirical settings, indicating that the estimated effects are not driven by model choice or sample selection. To address concerns about pre-existing differences in gender-related outcomes, I use a set of baseline gender indicators—such as the female labor force participation rate and the sex ratio among children aged 0 to 4—from earlier census waves, when only a limited number of Juancan had been established. These measures are used to assess whether the subsequent presence of Juancan is orthogonal to pre-existing gender outcomes.

There is a growing body of literature exploring the impact of immigrants on the behavior of local natives’ gender attitude. [Miho et al. \(2023\)](#) utilized the historical events from Stalin’s ethnic deportations and provided valuable insights into the profound connection between forced migration and the shift in local gender norms. These deportations led different ethnic groups

to redefine gender roles in new cultural environments, triggering the diffusion and integration of cultural norms. In particular, the policies promoting female labor force participation under the Soviet regime reflected the profound impact of gender-equal policies, which continue to influence local societies and cultures today.

Additionally, [Jessen et al. \(2024\)](#) studied the effects of East German immigration on the evolution of cultural norms and female labor supply in post-reunification Germany. They found that the more gender-egalitarian culture of East Germany had a lasting impact on the labor market behavior of West German women after the migration of East Germans. Specifically, the influx of East German migrants led to an increase in female labor force participation in the regions where they settled, driven largely by local cultural learning mechanisms rather than solely by economic factors or policy changes. Their study shows that the gender attitudes of immigrants not only affect the labor market participation of women in the receiving communities but also propagate through social interactions and community exposure. Similarly, [Boelmann et al. \(2025\)](#) further explored the interplay between childhood socialization and adulthood environment, highlighting the persistence of cultural norms in shaping maternal labor supply. Their research reveals that the gender-equal culture of East German women persisted even after they moved to the more gender-traditional West Germany, and they continued to display higher rates of labor force participation. This phenomenon points to the persistence of cultural norms and demonstrates the process of cultural adoption, where women gradually internalize the more egalitarian gender norms when exposed to them over time, even if they eventually return to their original cultural environments.

This paper contributes to the literature in several important ways. First, I exploit precise geographic information on the locations of public settlements (Juancan), which allows me to construct a more robust and spatially refined measure of exposure at a granular level. Second, the paper sheds light on the effects of hosting communities' exposure to a migrant population that was, on average, more highly educated and socioeconomically advantaged than the local population. This feature distinguishes the setting from much of the existing migration literature, which largely focuses on the impacts of hosting displaced or disadvantaged groups. Finally, by studying postwar

Taiwan, this paper provides new evidence from an East Asian context, complementing a literature that has been predominantly based on European or other regional experiences.

The remainder of this paper is organized as follows. Section 2 introduces the historical background, Section 3 describes the data and sample construction, Section 4 outlines the empirical strategy, Section 5 presents the main results, and Section 6 concludes.

2 Historical Background

After Japan’s surrender in 1945, Taiwan experienced a fundamental political transition. Japanese colonial authorities and residents departed the island, ending five decades of colonial rule. Administrative control was transferred to the Kuomintang-led Republic of China government, which took over governance in October 1945. The handover involved not only a change in political sovereignty but also a broad reorganization of state institutions, as the incoming government sought to replace Japanese colonial systems with those of the Nationalist regime.

The political transition after 1945 was soon followed by a large influx of migrants from mainland China. Between 1945 and 1950, the failure in the Chinese Civil War led to the relocation of a substantial number of military personnel, government officials, and civilians to Taiwan alongside the Kuomintang regime. This migration wave fundamentally changed Taiwan’s demographic composition and social structure, introducing a population that differed markedly from the local Taiwanese in terms of linguistic background, social norms, and prior institutional experience. Unlike earlier population movements, this inflow was closely tied to state-building, shaping the postwar political and economic landscape of Taiwan.

2.1 Gender Attitudes among Mainlanders and Taiwanese

There exist substantial disparities in gender attitudes between Mainlanders and local Taiwanese. From an institutional perspective, Mainlanders were disproportionately drawn from urban elites, military personnel, and their families—who possessed higher levels of education and greater exposure to modern intellectual movements. In particular, many of these elites were influenced by

the May Fourth Movement, which challenged Confucian patriarchal norms and advocated Western ideals such as individual autonomy, scientific rationality, and women’s liberation. This ideological shift among the elite facilitated a reconfiguration of gender norms, leading to more egalitarian attitudes toward female employment and social participation relative to those prevalent among local Taiwanese.

Figures 1 and 2 illustrate the main patterns in the census data along two dimensions: employment outcomes and son preference. In Figure 1, I compute employment rates for individuals aged 20 to 50 who are not students, disaggregated by ethnicity and gender, and further stratified by educational attainment, measured by whether an individual holds a high school degree. Panel (a) presents results for the full sample. Clear gender differences in employment outcomes emerge. On average, Mainlanders exhibit lower employment rates among men and among women with lower levels of education. In contrast, among highly educated women, Mainland women are more likely to be employed than their Taiwanese counterparts. Previous studies suggest that marriage constitutes a key constraint underlying gender gaps in labor market outcomes. Motivated by this insight, Panel (b) restricts the sample to married individuals. Among men, employment rates remain high for both Taiwanese and Mainlanders, with little difference across groups. Strikingly, among highly educated women, Mainland women appear substantially less affected by marriage, whereas Taiwanese women experience a pronounced decline in employment of approximately 13 percentage points. This pattern suggests that gender norms related to marriage and labor force participation differ markedly across ethnic groups.

Figure 2 presents patterns of behavioral son preference across ethnic groups. Son preference is measured by the probability of having a third child among families who already have two daughters, a commonly used indicator in the literature. The figure shows a substantial gap between Taiwanese and Mainlander families. In both survey years, Taiwanese families exhibit a significantly higher likelihood of having a third child following two daughters, indicating stronger son preference. Moreover, while son preference declines over time for both groups, the gap between Taiwanese and Mainlander households remains sizable. These patterns point to persistent differences in

gender norms related to fertility decisions across ethnic groups.

Furthermore, I use survey data to provide direct evidence on gender attitudes among Taiwanese and Mainlanders. Figure 3 reports the proportion of individuals who oppose specific statements related to the mother-in-law–daughter-in-law relationship and women’s political participation. A higher proportion indicates more gender-progressive attitudes. Across both dimensions shown in Panel (a) and Panel (b), Mainlanders consistently exhibit more gender-progressive views than their Taiwanese counterparts. This gap is particularly notable among women, suggesting substantial cross-ethnic differences in gender norms.

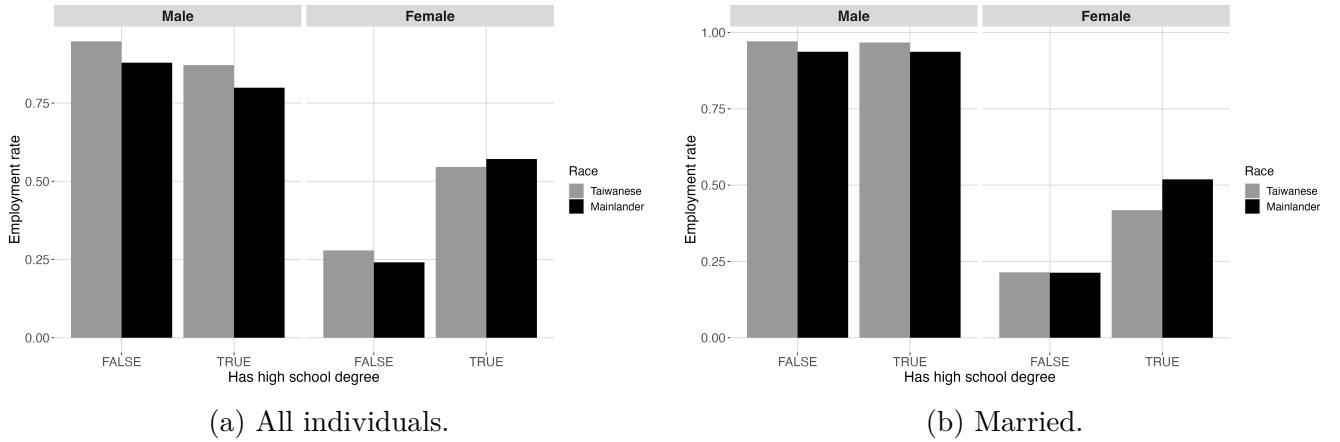


Figure 1: Employment rate by education and race

Notes: This figure plots employment rates by educational attainment and race. Panel A reports results for all individuals, while Panel B restricts the sample to married individuals. Bars represent mean employment rates within each group. Data source: 1980 Census.

2.2 Construction of Juancan

To accommodate the large inflow of Mainlanders, the government started to establish public settlements from 1940s. These public settlements were also known as *Juancan*. Figure 4 and 5 present the construction time and the geography distribution of the Juancan around 1990. In terms of geography, the settlements were primarily located in populous areas—including Taipei, Taoyuan, and Kaohsiung. In terms of timing, most Juancan were constructed in the 1950s and 1960s, well before the census years analyzed (1980 and 1990). As a result, their establishment does not overlap with the period of analysis.

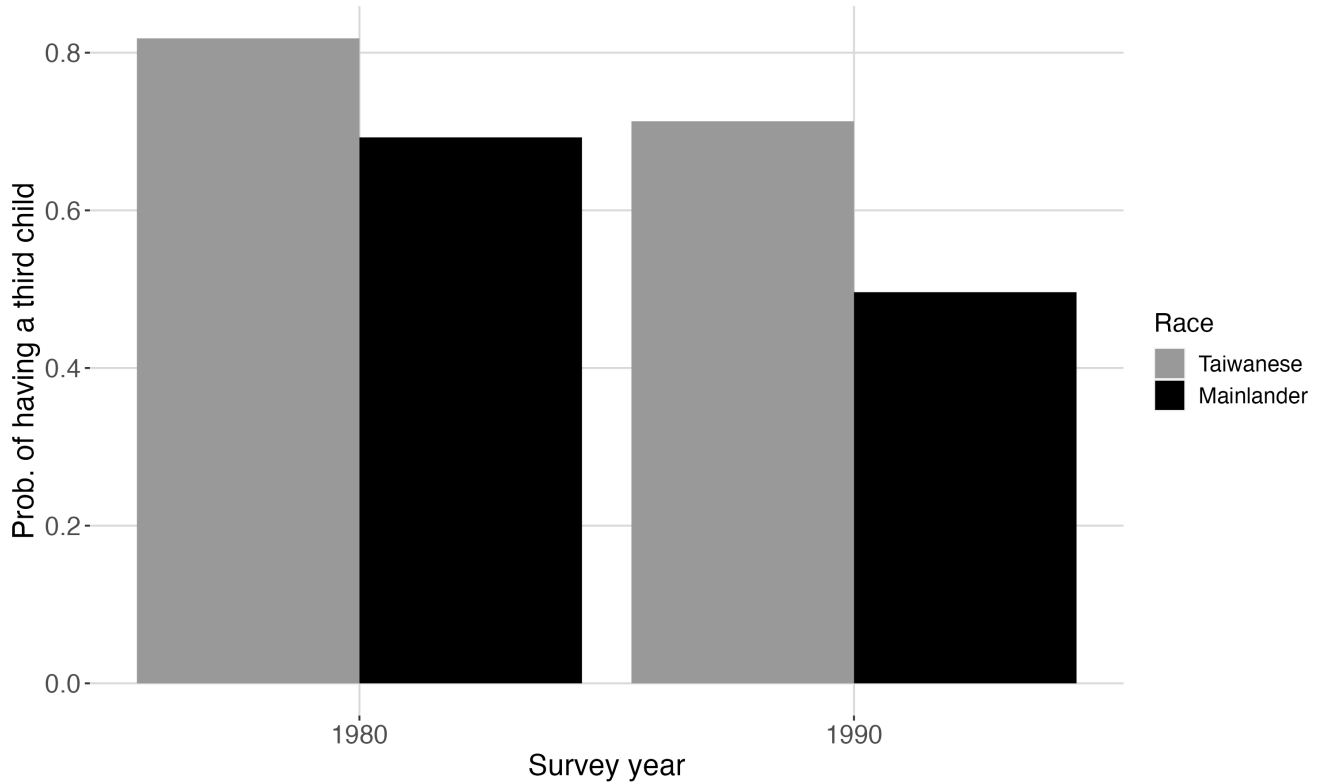
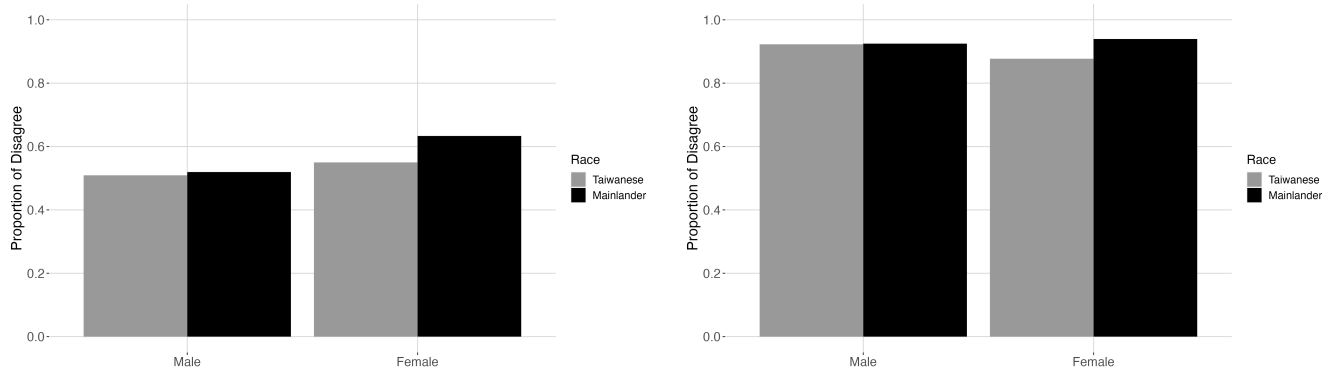


Figure 2: Son Preference

Notes: The dependent variable is the probability of having a third child conditional on already having two daughters. Data source: 1980 and 1990 Census.



(a) Mother-in-law and daughter-in-law relationship

(b) Women's political participation

Figure 3: Gender attitude

Notes: The figure reports disagreement rates with respect to specific statements related to gender attitudes. The original survey responses include four categories, "strongly agree", "slightly agree", "slightly disagree", and "strongly disagree". I recode these responses into a binary indicator that equals one if the respondent expresses any level of disagreement and zero otherwise. Panel (a) refers to the statement: "When a mother-in-law and a daughter-in-law come into conflict, even if the mother-in-law is in the wrong, the husband should encourage his wife to defer to the mother-in-law." Panel (b) refers to the statement: "Women should not be involved in politics as much as men." Data source: Barometer Survey.

The construction of Juancan was shaped by the urgent need to accommodate a large influx of military personnel and their families in the immediate postwar period. Early Juancan were largely formed through the reuse of Japanese military dormitories or through improvised housing arrangements, including the conversion of warehouses, schools, and other existing structures. When such facilities were insufficient, temporary dwellings were constructed using simple materials such as wood, bamboo, and metal sheets, reflecting the provisional nature of early settlements. These early Juancan were characterized by minimal planning and basic living conditions, with limited infrastructure and public amenities. Importantly, they were generally not enclosed by walls or fences and were often spatially integrated with surrounding urban or rural neighborhoods. Beginning in the late 1950s, housing construction became more standardized and institutionalized, with the gradual introduction of brick, stone, and reinforced concrete structures. Despite this transition, Juancan remained primarily functional settlements designed for accommodation rather than formal segregation, preserving a relatively open spatial relationship with the local population.

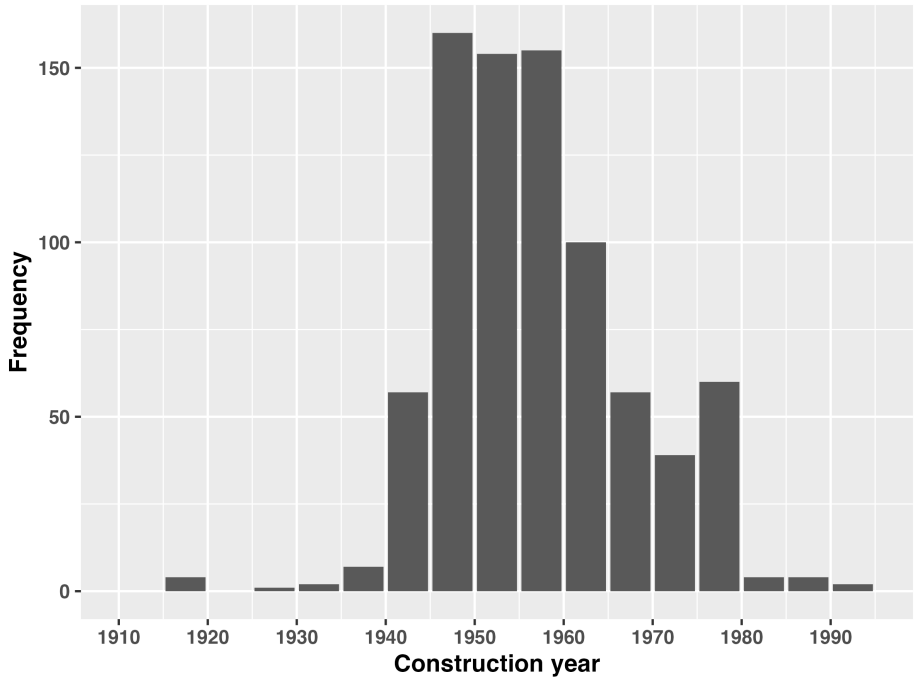


Figure 4: Construction year of Juancan.

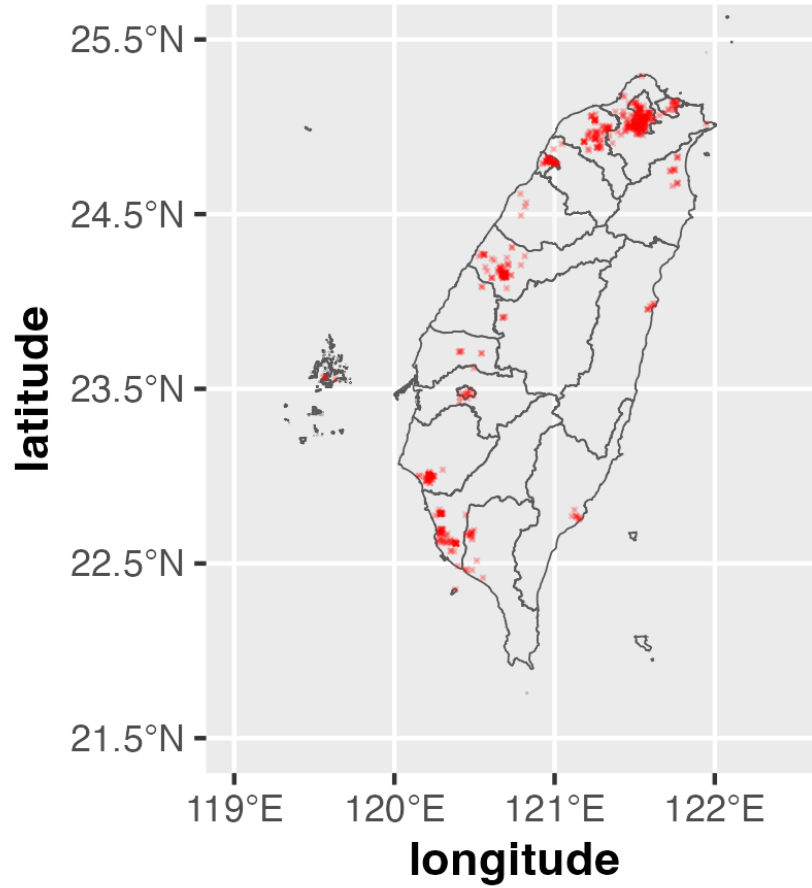


Figure 5: Geographical distribution of Juanca.

Notes: The figure shows the geographical distribution of Juanca, with darker shades indicating areas of higher concentration. As shown in the figure, the geographical distribution was highly concentrated in populous areas.

3 Data and Sample

This section will introduce the datasets I used and the following sample construction procedure.

3.1 Data

To examine the impact of Juanca on local Taiwanese's gender outcomes, I merged across various datasets, including census data, Juanca information data and contemporary geographical boundary data.

A. Population and Housing Census

The 1980 and 1990 Taiwan censuses are full enumerations of all households, offering rich information on individual and household characteristics. These data report each member’s gender, age, nationality, marital status, educational attainment, employment status, and place of residence. I focus on three dimensions of gender-related outcomes: female employment, fertility, and son preference. The individual-level census records allow me to define employment status precisely and to observe each household’s contemporary family composition, including the parity and sex of children. In addition, because the census is collected at the village level, it enables a more accurate measurement of local exposure to Juancan.

B. Juancan dataset

Information on military villages is primarily obtained from (Kuo, 2005). This dataset records all military villages built in Taiwan, along with their construction dates, total number of housing units, and their exact address.

C. 2017 Administrative Boundary Information

Due to the unavailability of administrative boundary data from 1980 and 1990, I mapped historical administrative units to their 2017 counterparts. Owing to changes in administrative boundaries over time, a full and precise match was not always feasible. The final matching rate was approximately 75%. After obtaining geographic coordinates for each administrative village, I calculated the distance to the each Juancan to further quantify exposure intensity.

3.2 Sample

To obtain comparable estimates of the effect of Juancan on natives’ gender-related outcomes, I construct the full sample using native individuals aged 20 to 50 who were not students in the 1980 and 1990 censuses. The fertility sample further restricts the data to married Taiwanese women who had at least one child in 1980 or 1990.

For the analysis of son preference, I impose an additional restriction and focus on individuals with at least two children in the survey year. Moreover, the second child must be younger than five years old. Because migration histories are only observed for the five years preceding the census, families with older children may have moved into or out of Juancan long before the observation window. In such cases, we might incorrectly attribute their gender-related outcomes to Juancan, even though these families may simply have had more progressive gender norms to begin with.

3.3 Construction of Exposure Intensity

To quantify the magnitude of the exposure effect, I construct an index that measures the cultural influence of Juancan j on individuals residing in village v . Specifically, I use GIS techniques to calculate the geographic distance between the Juancan j and the centroid of village v . To account for heterogeneity in the size of military villages, I incorporate their housing units into the index. The resulting measure takes the form

$$Exp_{jv} = Size_j / Dis_{jv}$$

, where a larger Juancan or a shorter distance implies greater potential exposure. The underlying intuition is that the size of a Juancan affects exposure linearly, while the effect of distance diminishes nonlinearly with increasing separation.

Given the state of transportation development in Taiwan during the 1980s and 1990s, I assume that individuals could effectively interact and exchange information within a 2-kilometer commuting zone. Accordingly, the aggregated exposure for individuals residing in village v is defined as

$$Exp_v = \sum_{j \in J} Exp_{jv} \times \mathbf{1}\{Dis_{jv} \leq 2 \text{ (km)}\},$$

where J denotes the set of all Juancan.

3.4 Summary Statistics

Table 1 reports summary statistics for the main variables used in the analysis, separately for female and male natives aged 20 to 50 in the 1980 and 1990 censuses. The results reveal a pronounced gender gap in employment rates: while over 90 percent of males are employed, only 40 percent of females participate in the labor market.

On average, the exposure intensity index is relatively small (mean = 0.004), but exhibits considerable variation across villages, suggesting spatial heterogeneity in cultural diffusion from nearby military villages. Education levels also differ by gender, with men more likely to hold a high school or higher degree. At the village level, the sex ratio averages around 1.06, implying slightly more men than women, and the average population size is about 8.1 in log scale. Approximately 67 percent of respondents lived in the same location five years earlier, indicating limited geographic mobility during this period.

Overall, these descriptive statistics highlight significant gender disparities in labor outcomes and socioeconomic characteristics, motivating the subsequent regression analysis that examines whether exposure to a gender-egalitarian society attenuated traditional gender norms and increased female labor participation.

Table 1: Summary Statistics

Variable	Female Sample				Male Sample			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Employed	0.404	0.491	0	1	0.912	0.283	0	1
Exposure	0.004	0.009	0	0.310	0.004	0.009	0	0.310
Age	32.574	8.455	20	50	33.197	8.348	20	50
High school or above degree	0.326	0.469	0	1	0.409	0.492	0	1
Live in the same place 5 years ago	0.678	0.467	0	1	0.741	0.438	0	1
Sex ratio of the village	1.063	0.134	0.522	11.840	1.076	0.179	0.522	11.840
log(Population of the village)	8.112	0.661	3.555	9.762	8.066	0.670	3.807	9.762
Observations	6,208,558				6,539,504			

Notes: This table reports summary statistics for the main variables used in the analysis, separately for female and male Taiwanese who were 20 to 50 years old in the 1980 and 1990 censuses. Since the exposure measure in this study is continuous, observations with extreme values are excluded from the sample. Specifically, individuals with an exposure index above 4.

4 Empirical Strategy and Identification

4.1 Empirical Specifications

The study aims at estimating the effect of the exposure to Juanca on the natives' gender outcomes. To account for regional development differences, I control for township fixed effects, which is a strong assumptions and is necessary for identification. Additionally, I also control for time-variant characteristics on village level. I estimate the following two equations on a cross-section of individuals. The effect of exposure on employment and fertility are estimated by :

$$Y_{ij(i)kt} = \alpha_0 + \beta_1 Exposure_{j(i)t} + \alpha_1 X_{it} + \alpha_2 \mu_{j(i)t} + \lambda_{k,t} + \varepsilon_{ij(i)kt}. \quad (1)$$

And the effect of exposure on son preference is estimated by :

$$\begin{aligned} Y_{ij(i)kt} = & \alpha + \beta_1 Exposure_{j(i)t} + \beta_2 SexComb_{it} \\ & + \beta_3 (Exposure_{j(i)t} \times SexComb_{it}) + \alpha_1 X_{it} + \alpha_2 \mu_{j(i)t} + \lambda_{k,t} + \varepsilon_{ij(i)kt}. \end{aligned} \quad (2)$$

i indexes individuals located in village $j(i)$ of township k . *Exposure* denotes the aggregated exposure within 2 km of village $j(i)$ at time t . Y denotes the outcome variables, such as employment status, total number of children, and the indicator for having a third child, etc. X and μ represents respectively the individual characteristics such as age, the square term of age, education level and village characteristics, for example, the log of population size and industry labor share. Finally, the λ stands for the township-by-time fixed effects, which is the main control variables necessary for identification.

4.2 Identification Assumptions

The main identification assumption underlying Equations (1) and (2) is that, conditional on township-by-time fixed effects and observable individual and village characteristics, the spatial dis-

tribution of Juancan captured by the exposure index is orthogonal to any unobserved determinants of gender preference in the local population.

In other words, after accounting for local socioeconomic conditions, population size, and industry composition, the variation in exposure intensity across villages within the same township and year reflects exogenous differences in proximity to mainlander community rather than pre-existing trends in gender norms or labor market outcomes.

Essentially, this assumption is untestable since it involves the unobserved characteristics. However, I am able to show that the presence of Juancan is unrelated to pre-existing gender differences. To further examine the assumption, I conduct a pre-trend test using 1956 census data, which was collected at the township level. Since some townships already had Juancan prior to 1956, I exclude these townships from the analysis. Table 2 shows that, after conditioning on township type, population size, and county fixed effects, neither the female employment rate nor the age 0-4 sex ratio predicts the establishment of military villages. However, the inverse of the distance to the nearest military base is positively correlated with the establishment of these villages. These results support the assumption that the location of military villages was primarily determined by strategic and administrative considerations rather than gender attitudes.

Table 2: Pre-trend Test.

Model:	Has Juancan			log(Size of Juancan + 1)		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Independent Variables</i>						
FEP	0.0251 (0.1403)			0.1735 (0.7816)		
Sex ratio (0–4)		0.0017 (0.0021)			0.0089 (0.0125)	
1 / Distance to military base			0.3212** (0.1433)			1.1760* (0.6156)
<i>Fit Statistics</i>						
Observations	285	285	285	285	285	285
Adj. R ²	0.2211	0.2216	0.2374	0.2167	0.2171	0.2244
Mean of Dep. Var.	0.158	0.158	0.158	0.683	0.683	0.683

Notes: This table examines the relationship between pre-treatment outcomes and the presence of Juancan. The outcome variables include a binary indicator for whether a township has any Juancan (left panel) and the total number of Juancan housing units within a township (right panel). Townships in which Juancan already existed prior to 1956 are excluded from the analysis. The control variables include the logarithm of population size and township type. Standard errors, reported in parentheses, are clustered at the county level. Data Source: 1956 Census. (Township Level) Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5 Results

This section discusses the empirical results under specification 1 and 2. The main outcomes include the employment, fertility and son preference.

5.1 Employment

Table 3 presents the effects of exposure to Juancan on employment outcomes among local Taiwanese individuals aged 20 to 50. To examine gender heterogeneity, I estimate the same specification separately for female and male samples. Columns (1)–(3) report results for (1) the full sample, (2) a subsample restricted to individuals who have not migrated within the past five years, and (3) a further restricted sample consisting of married non-migrants.

Across specifications, exposure to Juancan is positively associated with female employment,

whereas the corresponding estimates for males are generally small or negative and often statistically insignificant. Moreover, the estimated effect becomes stronger in column (2) and (3), where movers are excluded from the sample. One possible explanation is that women’s migration during this period was primarily driven by marriage or employment, with relatively clear intentions that were largely unrelated to exposure to Juancan. Excluding movers therefore sharpens the estimated effect by focusing on individuals with more stable residential histories. The stronger effects among non-movers also underscore the importance of accumulated exposure over time.

Turning to the sample restricted to married non-movers, the estimated effects differ sharply by gender: the effect remains positive for women but becomes negative for men. This pattern suggests a weakening of the traditional male breadwinner norm, potentially inducing married women to enter the labor market.

Moreover, the implied magnitudes are economically meaningful. A one-standard-deviation increase in exposure (0.009) is associated with an absolute increase of approximately 0.001–0.003 in the probability of female employment, corresponding to a 0.4–0.9 percent increase relative to the female mean. Moving from zero exposure to the maximum observed exposure (0.310) implies an increase in female employment of roughly 4.7–9.6 percentage points.

Taken together, these patterns indicate that the positive labor-market response is concentrated among women and is more consistent with an attenuation of traditional gender norms than with a uniform labor-market shock.

5.2 Fertility

Next, I examine the effect of exposure to Juancan on fertility behavior among local Taiwanese women. Table 4 presents the corresponding results. I focus on two outcomes: total number of children and the number of children born in the past five years. The former captures women’s completed or cumulative fertility, reflecting long-run fertility responses, while the latter provides a measure of more recent fertility decisions. The odd-numbered columns, which use the full sample, serve as the baseline results, while the even-numbered columns restrict the sample to non-movers

Table 3: The effect on labor force participation.

Dependent Variable:	Employed					
	Female			Male		
Model:	(1)	(2)	(3)	(1)	(2)	(3)
Exposure	0.1805** (0.0813)	0.2241** (0.0919)	0.3227*** (0.1100)	0.0030 (0.0958)	-0.1274 (0.0991)	-0.1860** (0.0865)
<i>Sample restriction</i>						
Non-mover		✓	✓		✓	✓
Married			✓			✓
Observations	6,208,558	4,206,996	3,090,386	6,539,504	4,843,086	3,110,678
Adjusted R ²	0.11789	0.12852	0.11043	0.07267	0.07408	0.01864
Mean of Dep. Var.	0.40428	0.41702	0.31355	0.91236	0.91264	0.96031

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education level, village industrial composition, village population size, and township-by-year fixed effects. Data source: 1980 and 1990 Census.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

in order to mitigate concerns about selective migration. Across all specifications, exposure to Juancan is associated with a statistically significant decline in fertility.

Columns (1) and (2) examine total number of children. The results show that greater exposure is associated with lower cumulative fertility, with larger magnitudes when the sample is restricted to non-movers. The estimates imply that a one-standard-deviation increase in exposure (0.009) reduces the total number of children by approximately 0.008–0.009. While moving from zero exposure to the maximum observed exposure (around 0.3) corresponds to a decline of roughly 0.26–0.31 children (10 percent relative change compared to the mean of dependent variable), indicating economically meaningful effects on completed fertility.

Columns (3) and (4) focus on more recent fertility behavior, measured by the number of children born in the past five years. The estimates remain negative and statistically significant, suggesting that exposure also affects contemporaneous fertility decisions. A one-standard-deviation increase in exposure is associated with a reduction of approximately 0.002–0.003 births, while moving from zero exposure to the maximum observed exposure implies a decline of about 0.07–0.09 births (10 percent relative change compared to the mean of dependent variable) over the past five years.

Table 4: The effect on fertility.

Dependent Variable: Model:	Total child number		Child number in past 5 years	
	(1)	(2)	(3)	(4)
Exposure	-0.8700*** (0.1346)	-1.017*** (0.2299)	-0.2872*** (0.0632)	-0.2315** (0.0765)
<i>Sample restriction</i>				
Non-mover		✓		✓
Observations	2,069,665	1,274,206	2,069,665	1,274,206
Adjusted R ²	0.30208	0.25242	0.39386	0.43328
Mean of Dep. Var.	2.5818	2.7871	0.65042	0.52208

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education level, village industrial composition, village population size, and township-by-year fixed effects. Data source: 1980 and 1990 Census.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 Son Preference

Finally, I investigate the effect of exposure to Juan-can on son preference. Son preference is measured by the propensity to have a third child conditional on already having two children. Table 5 presents the results. Column (1) includes all mothers, while Column (2) further restricts the sample to non-movers. The sex composition Son–Son serves as the omitted comparison group.

The second through fourth rows show that, relative to the Son–Son baseline, families with other sex combinations of the first two children exhibit a significantly higher likelihood of having a third child, particularly those with two daughters. This pattern reflects the strong son preference among local Taiwanese families during the 1980 and 1990 censuses.

The interaction terms in the fifth through seventh rows indicate that greater exposure to Juan-can is associated with a weakening of son preference. Notably, the negative interaction effects are largest for families whose first two children are both daughters. For these Daughter–Daughter families, moving from zero exposure to the maximum observed exposure implies a reduction of ap-

Table 5: The effect on son preference.

Dependent Variable:	Has Third Child	
Model:	(1)	(2)
Exposure	0.0078 (0.0144)	0.0103 (0.0145)
Son-Daughter	0.0440*** (0.0020)	0.0473*** (0.0027)
Daughter-Son	0.0418*** (0.0020)	0.0481*** (0.0027)
Daughter-Daughter	0.1664*** (0.0028)	0.1799*** (0.0036)
Exposure $\times$ Son-Daughter	-0.0211 (0.0271)	0.0055 (0.0474)
Exposure $\times$ Daughter-Son	-0.0469** (0.0191)	-0.0750*** (0.0248)
Exposure $\times$ Daughter-Daughter	-0.0936** (0.0407)	-0.1254* (0.0692)
<i>Sample restriction</i>		
Non-mover		✓
Observations	584,645	278,656
Adjusted R ²	0.16095	0.18792
Mean of Dep. Var.	0.30122	0.33520

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education level, village industrial composition, village population size, and township-by-year fixed effects. Data source: 1980 and 1990 Census.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

proximately 3 percentage points in the probability of having a third child, corresponding to about a 20 percent decline relative to the mean. This suggests that exposure to Juancan substantially attenuates son-preferring fertility behavior.

Taken together, the results across employment, fertility, and son preference point to a consistent pattern: exposure to Juancan primarily affected women’s economic and family decisions in ways that are consistent with a weakening of traditional gender norms. While female labor force participation increases with exposure, male employment responses are small or negative. At the same time, fertility declines both in terms of completed fertility and recent childbearing, and son-preferring fertility behavior is substantially weakened, especially among families with two daughters. The fact that these effects are stronger among non-movers further highlights the role of accumulated exposure over time. Overall, the evidence suggests that Juancan exposure reshaped household behavior through changes in gender roles and norms rather than through a uniform labor-market or fertility shock.

6 Conclusion

This paper studies the horizontal transmission of gender norms across ethnic groups by exploiting a unique historical episode in postwar Taiwan. Following the large-scale migration of mainland Chinese after the Chinese Civil War, the government constructed public settlements (Juancan) to accommodate incoming migrants who, on average, held more gender-egalitarian attitudes than the local Taiwanese population. Using precise geographic information on the locations and sizes of Juancan, this paper constructs a distance-based exposure measure and examines how local exposure shaped local Taiwanese gender-related outcomes.

The empirical results point to a consistent pattern across multiple dimensions. Greater exposure to Juancan is associated with higher female labor force participation, lower fertility, and a substantial weakening of son-preference-driven fertility behavior. These effects are concentrated among women, while employment responses for men are zero or negative, suggesting a shift in intra-household roles rather than a uniform labor market shock. The results are robust across al-

ternative specifications, distance thresholds, and sample restrictions, and pre-trend tests indicate that the placement of Juancan is orthogonal to pre-existing gender-related outcomes.

Taken together, the findings provide evidence that sustained local exposure to migrant communities with more gender-egalitarian norms can reshape economic behavior and family decisions in host communities. More broadly, this paper contributes to the literature on cultural diffusion by highlighting a setting in which migrants were not socially or economically marginalized, but instead occupied relatively influential positions in the postwar state-building process. By focusing on Taiwan, the analysis also adds new evidence from an East Asian context, complementing a literature that has largely focused on European migration episodes.

At the same time, this study has several limitations. First, while the empirical patterns are consistent with a cultural transmission mechanism, the analysis cannot fully disentangle the specific channels through which exposure affects behavior. In particular, census data do not allow direct observation of attitudes, beliefs, or intra-household bargaining processes. Second, although extensive controls and fixed effects mitigate concerns about endogenous settlement patterns, causal interpretation still relies on the assumption that unobserved local factors are not correlated with exposure intensity. Future research could address these limitations in several ways. One promising direction is to combine historical exposure measures with survey data on attitudes and social interactions, which would allow for a more direct examination of the mechanisms underlying cultural transmission. Survey-based evidence could help distinguish between channels such as social learning, peer effects, and changes in household bargaining power. More generally, linking historical migration shocks to contemporary attitudinal data may offer deeper insights into how gender norms evolve and persist over time

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Appendix

A.1 Robustness Checks

Table A.1: The effect on employed rate. (Continuous version)

Dependent Variable:	Employed					
	Female			Male		
Model:	(1)Full	(2)Stayer	(3)Married + Stayer	(1)Full	(2)Stayer	(3)Married + Stayer
Threshold: 1 km						
Exposure	0.2036** (0.0821)	0.2533*** (0.0880)	0.3661*** (0.1093)	0.1107 (0.0987)	-0.1120 (0.0988)	-0.1882** (0.0944)
Threshold: 1.5 km						
Exposure	0.1921** (0.0789)	0.2430*** (0.0886)	0.3366*** (0.1080)	0.0343 (0.0911)	-0.1052 (0.0950)	-0.1754** (0.0858)
Threshold: 2 km						
Exposure	0.1805** (0.0813)	0.2241** (0.0919)	0.3227*** (0.1100)	0.0030 (0.0958)	-0.1274 (0.0991)	-0.1860** (0.0865)
Threshold: 2.5 km						
Exposure	0.1962** (0.0840)	0.2419** (0.0972)	0.3421*** (0.1146)	-0.0597 (0.0989)	-0.1297 (0.1022)	-0.1783** (0.0867)
Threshold: 3 km						
Exposure	0.1710** (0.0864)	0.2171** (0.1015)	0.3171*** (0.1186)	-0.0485 (0.0955)	-0.1363 (0.1042)	-0.1831** (0.0878)
<i>Fit statistics</i>						
Observations	6,208,558	4,206,996	3,090,386	6,539,504	4,843,086	3,110,678
Mean of Dep. Var.	0.40428	0.41702	0.31355	0.91236	0.91264	0.96031

Notes: Clustered (township $\times$ year) standard errors in parentheses. Controls: age, age², education level, village industry labor share, village population size, and township-by-year fixed effects.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.2: The effect on employed rate. (Binary version)

Dependent Variable:	Employed					
	Female			Male		
Model:	(1)	(2)	(3)	(1)	(2)	(3)
Threshold: 1 km						
Has Juancun	0.0068*** (0.0019)	0.0085*** (0.0021)	0.0088*** (0.0026)	-0.0031 (0.0036)	-0.0021 (0.0014)	-0.0025*** (0.0009)
Threshold: 1.5 km						
Has Juancun	0.0049** (0.0024)	0.0074** (0.0029)	0.0069** (0.0034)	-0.0017 (0.0042)	-0.0019 (0.0019)	-0.0026** (0.0011)
Threshold: 2 km						
Has Juancun	0.0046 (0.0032)	0.0061* (0.0034)	0.0059 (0.0039)	-0.0009 (0.0035)	-0.0023 (0.0016)	-0.0025*** (0.0009)
Threshold: 2.5 km						
Has Juancun	0.0009 (0.0036)	0.0020 (0.0039)	0.0026 (0.0043)	-0.0079* (0.0047)	-0.0042** (0.0017)	-0.0031*** (0.0010)
Threshold: 3 km						
Has Juancun	-0.0013 (0.0037)	-0.0004 (0.0039)	4.4×10^{-5} (0.0044)	-0.0045 (0.0044)	-0.0031* (0.0016)	-0.0020** (0.0010)
<i>Fit statistics</i>						
Observations	6,208,558	4,206,996	3,090,386	6,539,504	4,843,086	3,110,678
Mean of Dep. Var.	0.40428	0.41702	0.31355	0.91236	0.91264	0.96031

Notes: Clustered (township $\times$ year) standard errors in parentheses. Controls: age, age², education level, village industry labor share, village population size, and township-by-year fixed effects. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.3: The effect on son preference

Dependent Variable:	Has Third Child			
	Continuous		Binary	
Model:	(1)	(2)	(3)	(4)
Exposure	0.0078 (0.0144)	0.0103 (0.0145)	0.0105*** (0.0034)	0.0114*** (0.0042)
firstTwoBorn_sex12	0.0440*** (0.0020)	0.0473*** (0.0027)	0.0570*** (0.0028)	0.0650*** (0.0035)
firstTwoBorn_sex21	0.0418*** (0.0020)	0.0481*** (0.0027)	0.0559*** (0.0029)	0.0620*** (0.0038)
firstTwoBorn_sex22	0.1664*** (0.0028)	0.1799*** (0.0036)	0.1819*** (0.0031)	0.1962*** (0.0041)
Exposure $\times$ firstTwoBorn_sex12	-0.0211 (0.0271)	0.0055 (0.0474)	-0.0216*** (0.0034)	-0.0329*** (0.0044)
Exposure $\times$ firstTwoBorn_sex21	-0.0469** (0.0191)	-0.0750*** (0.0248)	-0.0236*** (0.0035)	-0.0266*** (0.0047)
Exposure $\times$ firstTwoBorn_sex22	-0.0936** (0.0407)	-0.1254* (0.0692)	-0.0263*** (0.0047)	-0.0316*** (0.0059)
<i>Fit statistics</i>				
Observations	584,645	278,656	584,645	278,656
Adjusted R ²	0.16095	0.18792	0.16107	0.18813
Mean of Dep. Var.	0.30122	0.33520	0.30122	0.33520

Notes: Clustered (township $\times$ year) standard errors in parentheses. Controls: age, age², education level, village industry labor share, village population size, and township-by-year fixed effects.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.4: The effect on son preference - 1 km

Dependent Variable:	Has Third Child			
	Continuous		Binary	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Exposure (exposure1_agg)	-0.0020 (0.0087)	-0.0008 (0.0080)	0.0097*** (0.0031)	0.0162*** (0.0038)
firstTwoBorn_sex12	0.0439*** (0.0020)	0.0472*** (0.0027)	0.0511*** (0.0025)	0.0570*** (0.0032)
firstTwoBorn_sex21	0.0417*** (0.0020)	0.0479*** (0.0027)	0.0490*** (0.0024)	0.0565*** (0.0033)
firstTwoBorn_sex22	0.1661*** (0.0028)	0.1796*** (0.0036)	0.1754*** (0.0027)	0.1913*** (0.0037)
Exposure $\times$ firstTwoBorn_sex12	-0.0074 (0.0197)	0.0259 (0.0389)	-0.0176*** (0.0032)	-0.0265*** (0.0046)
Exposure $\times$ firstTwoBorn_sex21	-0.0345*** (0.0121)	-0.0630*** (0.0179)	-0.0181*** (0.0032)	-0.0238*** (0.0044)
Exposure $\times$ firstTwoBorn_sex22	-0.0731*** (0.0276)	-0.0994** (0.0500)	-0.0232*** (0.0047)	-0.0328*** (0.0059)
<i>Fit statistics</i>				
Dependent variable mean	0.30122	0.33520	0.30122	0.33520
Observations	584,645	278,656	584,645	278,656
Adjusted R ²	0.16094	0.18791	0.16103	0.18806

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education level, village industrial composition, village population size, and township-by-year fixed effects. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.5: The effect on son preference - 1.5 km

Dependent Variable:	Has Third Child			
	Continuous		Binary	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Exposure (exposure1_agg)	0.0026 (0.0114)	0.0041 (0.0109)	0.0140*** (0.0033)	0.0165*** (0.0041)
firstTwoBorn_sex12	0.0439*** (0.0020)	0.0472*** (0.0027)	0.0546*** (0.0026)	0.0617*** (0.0034)
firstTwoBorn_sex21	0.0417*** (0.0020)	0.0480*** (0.0027)	0.0533*** (0.0027)	0.0609*** (0.0036)
firstTwoBorn_sex22	0.1662*** (0.0028)	0.1797*** (0.0036)	0.1794*** (0.0030)	0.1937*** (0.0040)
Exposure $\times$ firstTwoBorn_sex12	-0.0134 (0.0229)	0.0178 (0.0418)	-0.0202*** (0.0031)	-0.0305*** (0.0042)
Exposure $\times$ firstTwoBorn_sex21	-0.0404*** (0.0154)	-0.0682*** (0.0209)	-0.0220*** (0.0033)	-0.0279*** (0.0044)
Exposure $\times$ firstTwoBorn_sex22	-0.0832** (0.0340)	-0.1112* (0.0587)	-0.0253*** (0.0046)	-0.0304*** (0.0060)
<i>Fit statistics</i>				
Dependent variable mean	0.30122	0.33520	0.30122	0.33520
Observations	584,645	278,656	584,645	278,656
Adjusted R ²	0.16094	0.18791	0.16105	0.18809

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education level, village industrial composition, village population size, and township-by-year fixed effects. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.6: The effect on son preference - 2.5 km

Dependent Variable:	Has Third Child			
	Continuous		Binary	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Exposure (exposure1_agg)	0.0123 (0.0172)	0.0146 (0.0174)	0.0100*** (0.0038)	0.0090* (0.0049)
firstTwoBorn_sex12	0.0440*** (0.0020)	0.0473*** (0.0027)	0.0588*** (0.0029)	0.0655*** (0.0038)
firstTwoBorn_sex21	0.0419*** (0.0020)	0.0482*** (0.0027)	0.0584*** (0.0030)	0.0641*** (0.0039)
firstTwoBorn_sex22	0.1665*** (0.0028)	0.1801*** (0.0036)	0.1837*** (0.0032)	0.1954*** (0.0042)
Exposure $\times$ firstTwoBorn_sex12	-0.0273 (0.0306)	-0.0036 (0.0523)	-0.0228*** (0.0037)	-0.0313*** (0.0047)
Exposure $\times$ firstTwoBorn_sex21	-0.0521** (0.0222)	-0.0793*** (0.0275)	-0.0255*** (0.0036)	-0.0283*** (0.0049)
Exposure $\times$ firstTwoBorn_sex22	-0.1022** (0.0461)	-0.1345* (0.0759)	-0.0270*** (0.0047)	-0.0277*** (0.0062)
<i>Fit statistics</i>				
Dependent variable mean	0.30122	0.33520	0.30122	0.33520
Observations	584,645	278,656	584,645	278,656
Adjusted R ²	0.16095	0.18792	0.16108	0.18811

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education level, village industrial composition, village population size, and township-by-year fixed effects. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.7: The effect on son preference - 3 km

Dependent Variable:	Has Third Child			
	Continuous		Binary	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Exposure (exposure1_agg)	0.0165 (0.0198)	0.0183 (0.0196)	0.0074* (0.0043)	0.0085 (0.0055)
firstTwoBorn_sex12	0.0441*** (0.0020)	0.0474*** (0.0027)	0.0615*** (0.0032)	0.0682*** (0.0040)
firstTwoBorn_sex21	0.0420*** (0.0020)	0.0483*** (0.0027)	0.0595*** (0.0032)	0.0659*** (0.0041)
firstTwoBorn_sex22	0.1667*** (0.0028)	0.1803*** (0.0036)	0.1839*** (0.0034)	0.1944*** (0.0044)
Exposure $\times$ firstTwoBorn_sex12	-0.0335 (0.0341)	-0.0124 (0.0569)	-0.0251*** (0.0039)	-0.0335*** (0.0050)
Exposure $\times$ firstTwoBorn_sex21	-0.0573** (0.0254)	-0.0850*** (0.0310)	-0.0255*** (0.0039)	-0.0291*** (0.0050)
Exposure $\times$ firstTwoBorn_sex22	-0.1098** (0.0512)	-0.1453* (0.0843)	-0.0256*** (0.0048)	-0.0243*** (0.0064)
<i>Fit statistics</i>				
Dependent variable mean	0.30122	0.33520	0.30122	0.33520
Observations	584,645	278,656	584,645	278,656
Adjusted R ²	0.16095	0.18792	0.16108	0.18811

Notes: Standard errors clustered at the township-by-year level shown in parentheses. Controls include age, age², education, village industrial composition, village population, and township-by-year fixed effects. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.